

Perplexity

How surprising are
the observed tokens?

Schematic token probabilities



Less likely means more surprise.

$$\text{PPL} = \exp(\text{mean surprise})$$

Surprise = negative log probability

Binoculars

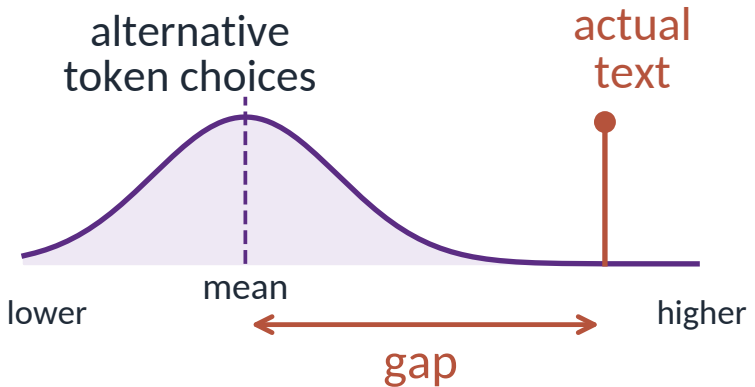
log-PPL / cross-entropy
from two related models

Word labels shown; models operate on tokens.

Fast-DetectGPT

Conditional probability curvature

Compare actual choices with alternatives.



Log-probability score

$$\frac{\text{actual score} - \text{expected score}}{\text{SD of alternative scores}}$$

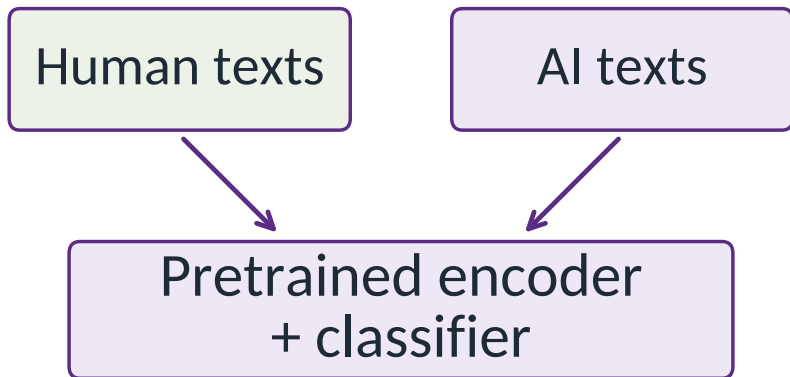
Keep each original prefix fixed.

A larger standardized gap is
a stronger machine-text signal.

Schematic distribution; no model outputs shown.

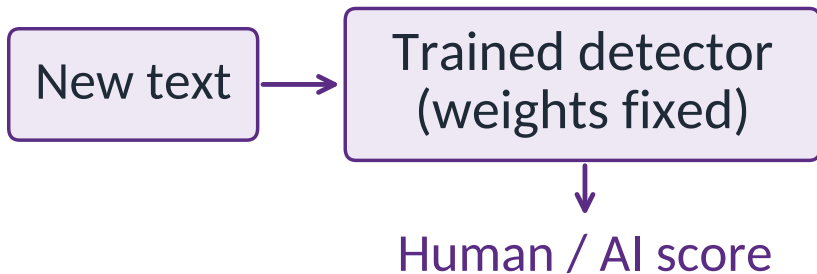
Fine-tuned RoBERTa

Learn from labelled examples



Update weights to predict the labels

Then score an unseen text



Supervised classification